

Mobile Car Detailing Scheduler

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CS 390 Project Milestone 1: Project Proposal

Abstract

Evansville Mobile Car Detailing is a new business, owned by Lily's brother Carson, with a rapidly growing clientele. Currently, appointments are set up by the owner manually through phone calls, texts, various social media messages, and word of mouth, which requires the owner to manually collect information, calculate the time needed for each job, and put each job on a Google Calendar. However, this process is becoming highly inefficient as his business grows. This project proposes a simple web-based scheduling system designed specifically for the needs of a mobile car detailing business. Clients would be able to provide their information, select detailing services and add-ons, and choose from available appointment times. The system would automatically calculate the estimated time needed for each job, while accounting for the selected services and the travel time. This project also leaves room for expansion to include reminders, payments, and no-shows fees if time allows. The goal is to reduce the amount of manual scheduling required by the owner, while also creating a simple, reliable, and professional scheduling experience for clients.

Current Practice

The current scheduling system for Evansville Mobile Car Detailing is through phone calls, texts, messages via social media (facebook and instagram), and word of mouth. The owner has to keep up with messages in multiple places, communicate back and forth with each client to collect all the needed contact information, vehicle info, desired services, location, and time. This information is manually collected and used to calculate how long the job should take, which depends on both the services selected and how long the commute is. The appointment is manually put on a Google Calendar.

Problem

Evansville Mobile Car Detailing is a new business, with a rapidly growing clientele. As the number of clients increases, keeping track of these various, scattered forms of communication is becoming increasingly difficult and time consuming. Scheduling requires more than just finding an open time. It must account for different detailing levels and add-ons, and, since the business is mobile, the location of each client and travel time between appointments also needs to be considered. Managing all of this manually takes time away from actual jobs and creates more opportunities for scheduling mistakes.

Solution

The owner has proposed a custom software that would automate the appointment scheduling, in a way that is simple and reliable enough to attract rather than deter clients. The proposed solution should have the ability for the owner to set available time slots, levels of detailing, and add-on options. The client should be able to efficiently provide contact information, car information, the desired detailing package, and any add-ons. Then, an automatic calculation of time needed per job, including the selected services and travel time (not just from the business address but from jobs that come before the time slot), would allow the client to select an adequate and efficient timeframe. Additional features could include appointment confirmation and reminders, and even payments with no-show fees.

Impact

While this project could be useful to detailing businesses in general, it will be tailored specifically to Evansville Mobile Car Detailing. A more uniform, reliable, and automated approach would result in less management stress, less scheduling mistakes, and a more professional client experience. Further, appointment confirmations and reminders could reduce no-shows and improve professionalism.

Technical Approach

Clients may be deterred by having to download an app. A simple link to a website would be standard, quick, and easy for clients to access from their phones or their computers.

We have both used React Native to design applications and websites. The scheduling system will also need a way to store appointment, service, and availability information so that the owner can manage the schedule and clients can see available appointment times, such as through supabase.

The website will require a dashboard for the owner to view appointments, manage his availability, and set up packages. The client side should be form like, with an emphasis on simplicity, carefully tailored questions, and strong error prevention.

Risks

Protection of personal information is a concern when names, addresses, and payment information is involved. The website should use a widely trusted third party for anything handling payment information or storage of personal information. Another concern is the potential to lose access to crucial appointment information should the website or the server be down. A potential solution is to have a company email specifically for backups, and the software could send a daily report of appointments as backup in case of a software failure.